

Lab #4: PID Control of Cart Position

Released: Feb 17, 2026

Due: Feb 18/19, 2026

Objectives

1. Learn how to derive the closed-loop transfer function for a system with a PID controller.
2. Design a PID controller that interfaces with the hardware using Simulink.
3. Gain hands-on experience with PID control of position of a cart on a track using the Quanser setup.
4. Compare and contrast the performance of different components of the PID control and understand how each component affects the response of the system.

1 Background

1.1 Modeling and control of the cart position

From the previous lab, we know that the transfer function of the linear velocity of the IP02 (Quanser) cart with respect to the input motor voltage is given by

$$\frac{V_{cart}(s)}{V_{motor}(s)} = \frac{K}{\tau s + 1} \quad (1)$$

where V_{cart} is the Laplace transform for the cart velocity $[v_{cart}(t)]$, V_{motor} is the Laplace transform for the motor input voltage $[v_m(t)]$, K is the steady-state gain, τ is the time constant and s is the Laplace operator.

From lectures, we know that controllers can be designed to tune the performance of systems. A specific controller that we will consider in this lab is the Proportional-Integral-Derivative (PID) controller. The PID controller has three components: a

proportional term, an integral term, and a derivative term. Each of these terms has a gain associated with it, which we can tune to achieve our desired performance specifications. The proportional term is based on the current error, the integral term is based on the accumulation of past errors, and the derivative term is based on the prediction of future errors. Mathematically, we have the following control law for the PID controller:

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau) d\tau + K_d \frac{de(t)}{dt} \quad (2)$$

where $u(t)$ is the control input, $e(t)$ is the error between the desired output and the actual output, K_p is the proportional gain, K_i is the integral gain, and K_d is the derivative gain.

Pre-Lab Quiz 1.1: Make sure you're ready for the lab!

Derive the transfer function of a PID controller in the Laplace domain.

Hint: What is the output of the controller? What is the input? The transfer function is the ratio of the output to the input in the Laplace domain.

We have the following block diagram for a system with a PID controller:

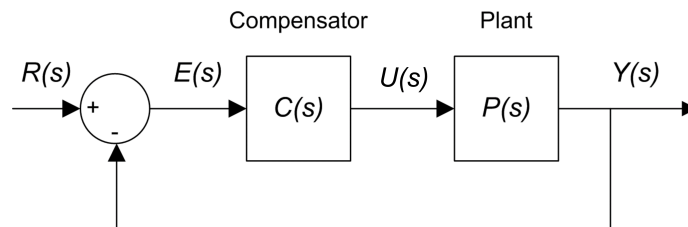


Figure 1: Unity Feedback System

Then $R(s)$, $E(s)$, and $U(s)$ be the Laplace transforms of the reference signal $r(t)$, error $e(t) = r(t) - y(t)$, and the input to the plant $u(t)$. Here, $r = x_d(t)$ is the set-point (desired) cart position, $y = x_a(t)$ is the measured cart position, and $u = v_m(t)$ is the control input or IP02 motor input voltage. Note that the error is ($e = r - y$).

Pre-Lab Quiz 1.2: Make sure you're ready for the lab!

Find out the closed-loop transfer function of the system with the PID controller.

Remember that to get the voltage-to-position transfer function $P(s) = Y(s)/U(s)$, we introduced a integrator $\frac{1}{s}$ in series with the voltage-to-speed transfer function from equation (1) (effectively integrating the speed to get position). The result is

$$P(s) = \frac{Y(s)}{U(s)} = \frac{K}{s(\tau s + 1)} \quad (3)$$

2 Lab Goals: Desired Specifications

By now, you should already be familiar with certain performance metrics for control systems: the rise time, the steady-state error, and the percent overshoot. Refer to Lab 3 for detailed derivation of these metrics. The minimum desired time-domain specifications are:

- peak time, $t_p = 0.1s$
- percent overshoot, $PO = 10\%$
- steady-state error, $e_{ss} = 0$ (must be less than 0.01)

Percent overshoot is dependent on the damping ratio, ζ , and can be expressed as:

$$PO = 100 \exp \left\{ \left(\frac{-\pi\zeta}{\sqrt{1-\zeta^2}} \right) \right\} \quad (4)$$

The peak time, t_p , is given by

$$t_p = \frac{\pi}{\omega_n \sqrt{1-\zeta^2}} \quad (5)$$

Finally, the steady-state error, e_{ss} , can be computed using the following formula:

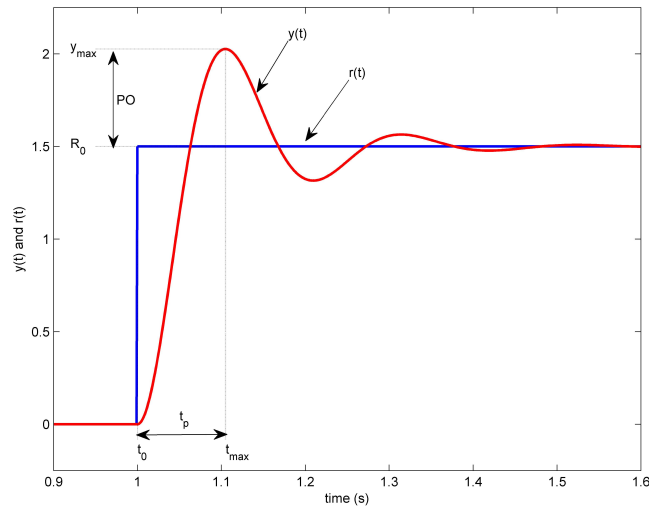


Figure 2: Standard Second-Order Step Response

$$e(t) = r(t) - y(t) \quad (6)$$

$$e_{ss} = \lim_{t \rightarrow \infty} e(t) = \lim_{s \rightarrow 0} sE(s) \quad (7)$$

Pre-Lab Quiz 2.1: Make sure you're ready for the lab!

Compute the theoretical values for the peak time, steady-state error, and percent overshoot for the closed-loop system with PID control by applying the formula above. Use the values of mass and τ from the previous questions and choose PID parameters arbitrarily (for now).

Pre-Lab Quiz 2.2: Make sure you're ready for the lab!

Show your TA that you have MATLAB and Simulink installed and running on your computer so that you are ready to perform the simulation steps below.

3 Experimental Steps: Individual Simulations

This experiment can only be performed on the cart position control setup. However, to save time on the equipment, you are asked to setup a simulation by building a Simulink model on your own computer (any MATLAB version is fine). You will then use the parameters from your simulation + use your understanding of how to create a controller in Simulink to run the experiment on the Quanser setup.

3.1 Building Simulink model and running the simulation

On your computer **individually**, create a PID controller in Simulink by following the steps below:

- a. Download the [lab4_model_setup_params.m](#) file from the course website and run it to get the parameters for the plant transfer function $P(s)$. The file also provides instructions for creating the Simulink model.
- b. Open Simulink and create a transfer function block

$$P(s) = \frac{K}{\tau s + 1}$$

using K and τ that should exist already in your MATLAB workspace.

- c. Insert the signal generator. Set to square wave input with frequency of 0.05 Hertz. This is your desired position $x_{\text{desired}}(t)$.
- d. Insert a subsystem block for the PID controller to mimic the way the model is setup in the Quanser equipment. Create two inputs (one for $x_{\text{desired}}(t)$ and one for $x_{\text{current}}(t)$) and one output (the control signal $u(t)$).
- e. Insert a Scope and a “Mux” block to read both the desired and current position signals.
- f. Connect output of the plant to the current position input of the PID controller (feedback signal) and to the Mux block.
- g. Connect the output of the PID controller to the input of the plant.
- h. Connect the desired position signal to the other input of the PID controller and to the Mux block.

Param	Rise time	Overshoot	Settling time	Steady state error	Stability
k_p	Decrease	Increase	Small change	Decrease	Degrade
k_i	Decrease	Increase	Increase	Eliminate	Degrade
k_d	Minor change	Decrease	Decrease	No effect in theory	Improve if k_d small

Table 1: Effects of increasing each PID parameter independently on the response of the system. Source: Wikipedia

- i. Connect the output of the Mux block to the Scope.
- j. Now, double click the subsystem block to create the PID controller by following the steps below.

3.2 Controller model in Simulink

- a. Compute the error signal $e(t) = x_{\text{desired}}(t) - x_{\text{current}}(t)$ using a sum block with the appropriate signs.
- b. Create the gain blocks for k_p , k_i and k_d . Make sure to change the names and the units for each gain block. To find the units, note that the control signal (output of controller) is voltage. Match up the units for the left hand side (u) with the units for the right hand side ($K_p e + K_i \int e + K_d \frac{de}{dt}$) to find the units for each gain block.
- c. To get integral action, insert an “Integrator” block and to get the derivative action, insert a “Derivative” block. When prompted for the value of “coefficient” (c), you can set it to 0.02 (this is a small value to prevent noise amplification — we will learn more about this later).
- d. Create a summer block that can add three signals together. The control output is the sum of the proportional, integral, and derivative terms.

Run your Simulink model and observe the response in the Scope. Adjust the values of k_p , k_i , and k_d to see how the response changes. Try to meet the desired specifications for the response by tuning the PID parameters.

Refer to Table 1 for the impact of each of the PID parameters on the response.

Pre-Lab Grade and Experimental Steps

Due: Feb 18/19, 2026 Name: _____

Grade: _____

1 Hands-on Experiment: Group Work

You can only proceed with these steps if **all members of the group** have finished the Simulink setup (individually) on their computer and have an initial set of values of the PID controller to try on the setup. The TA will check your simulations on your Simulink Scope. This requirement is **in addition to the pre-lab** work (answers to the quiz questions above).

On the lab computer, make sure that you change the workspace directory to the folder on the Desktop labeled “Lab #4”. Open ip02_quick_start on Simulink (see Figure 3 below).

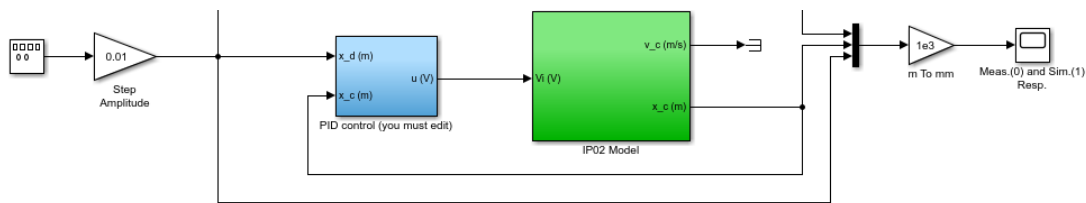


Figure 3: Simulink block diagram for position control simulation

1. Run the `setup_ip02.position` script
2. To generate a step reference, verify the **Signal Generator** is set to the following: Signal Type = **square**, Amplitude = 1, Frequency = 0.05 Hz.
3. Set the **Amplitude (m)** gain block to 0.01 to generate a step of 10 mm.
4. Double click the controller block to construct the controller by emulating the setup from your Simulink. Make sure that the **Control** subsystem implements a PID controller.
5. Open the cart position scope, x_c m and the motor input voltage scope, V_m V

6. Start the simulation and observe the response with the parameters assigned; take note of the colored lines in the x_c graph: **Yellow** Trace: set-point position
Purple Trace: simulated position
7. Set the values for k_p, k_i, k_d in the MATLAB workspace to the values you obtained in your simulations and observe how the changes affect the response. **DO NOT** set high values of k_d (must be less than 5). **IMPORTANT:** You only have three tries! For more data, return back when all groups are done. Set your parameter values to the values you obtained from your simulations. Then, run the experiment (see steps below for running the setup) to save the data.
8. Export the data from the MATLAB workspace variables to your personal computer to compute the steady-state error, percent overshoot, and peak time. Helpful hints on computation:

```

- x_c(m): data_pos
    * data_pos(:,1) = time
    * data_pos(:,2) = simulated position
    * data_pos(:,3) = set-point position
- data_vm (V): data_vm
    * data_vm(:,1) = time
    * data_vm(:,2) = simulated input voltage

```

1.1 Running the experimental setup

These steps are same as before. Open the Simulink file named `ip02_quick_start` to view the block diagram.

At the top, open the “Monitor & Tune” drop-down menu and click on “Build for Monitoring”

- Verify that the physical amplifier unit is turned on with the gain set to 1x
- Position the cart so that it is aligned in the center of the railing

Once the file has compiled, click on **Connect** then **Start**; click “**Stop**” when you see 3 steps on the graphs

- Keep clear of the rail the cart travels on until the connection has been stopped

Export the data from `data_xc` and `data_vm`

2 Action Items

In your lab report, provide the following:

1. Provide a table with the parameter values used for the position control (k_p, k_i, k_d)
2. Provide graphs of the collected data.
3. Calculate the steady-state error, the percent overshoot, and the peak time for the Position and Speed data.
4. Comment on the effect of each controller parameter.

PID parameters	e_{ss}	% PO	t_p (s)
1. $k_p =$ $k_i =$ $k_d =$			
2. $k_p =$ $k_i =$ $k_d =$			
3. $k_p =$ $k_i =$ $k_d =$			

Discuss your results in detail here.

Appendix: Relevant Files

File Name	Description
setup_ip02_position.m	Run this file only to set up the experiment's position control parameters.
config_ip02.m	Returns the configuration-based IP02 model specifications Rm , Jm , Kt , Eff_m , Km , Kg , Eff_g , M , r_{mp} , and Beq
d_ip02_position.m	Determines the response specifications ζ and ω_n , and the control gains k_p and k_v .
d_model_param.m	Determines the model parameters K , and τ
s_ip02_position.mdl	Simulink file that simulates the closed-loop IP02 position control step response
q_ip02_position.mdl	Simulink file that implements the closed-loop IP02 position controller using QUARC

Table 2: Matlab files needed for the Position Control Experiment

File Name	Description
setup_ip02_speed.m	Run this file only to set up the experiment's speed control parameters.
config_ip02.m	Returns the configuration-based IP02 model specifications Rm , Jm , Kt , Eff_m , Km , Kg , Eff_g , M , r_{mp} , and Beq
d_ip02_speed_lead.m	Determines the control parameters K_c and a , and T .
d_ip02_speed_pi.m	Determines the control parameters K_p and k_i .
d_model_param.m	Determines the PI control parameters K , and τ
s_ip02_speed_lead.mdl	Simulink file that simulates the closed-loop IP02 lead speed control step response
q_ip02_speed_lead.mdl	Simulink file that implements the closed-loop IP02 lead speed controller using QUARC
q_ip02_speed_pi	Simulink file that implements the closed-loop IP02 PI speed controller

Table 3: Matlab files needed for the Speed Control Experiment

Appendix: Equations of Motion Derivation

In this section, we derive the equations of motion for the cart system. By Newton's Second Law, we can represent the force between the cart's DC motor and the motion as follows:

$$M\dot{v}_c(t) = F_c(t) - B_c v_c(t) \quad (8)$$

where M is the mass of the cart, v_c is the linear velocity of the cart, and B_c is the equivalent viscous damping coefficient as seen at the motor pinion. The driving force generated by the motor on the cart can be expressed as:

$$F_c = \frac{\eta_g K_g \tau_m}{r_{mp}} \quad (9)$$

where η_g is the gearbox efficiency, K_g is the gear ratio, τ_m is the torque generated by the motor, and r_{mp} is the motor pinion radius.

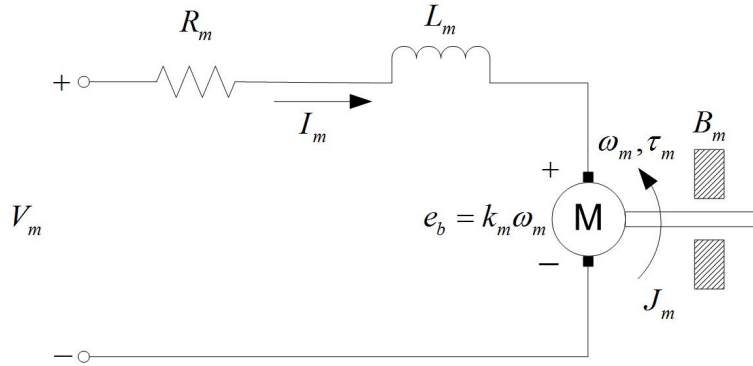


Figure 4: IP02 DC Motor Armature Circuit

Figure 4 displays the circuit for the driving motor where R_m is the motor resistance, L_m is the inductance, B_m is the motor damping, and k_m is the back-emf constant. Using Kirchoff's Voltage Law, we can write the voltage equation to solve for the inductance (given that L_m represents a much smaller value, it can be neglected):

$$V_m(t) - R_m I_m(t) - L_m \dot{I}_m(t) - k_m \omega_m(t) = 0 \rightarrow I_m(t) = \frac{V_m(t) - k_m \omega_m(t)}{R_m} \quad (10)$$

By introducing the motor torque constant, k_t , we can expand equation (3) by substituting all the components:

$$F_c = \frac{\eta_g K_g \eta_m k_t (V_m(t) - k_m \omega_m(t))}{R_m r_{mp}} \quad \text{where} \quad \tau_m(t) = \eta_m k_t I_m(t) \quad (11)$$

The translation between angular to linear velocity from the motor to the cart is as follows

$$F_c = \frac{\eta_g K_g \eta_m k_t (V_m(t) r_{mp} - k_m K_g v_c(t))}{r_{mp}^2 R_m} \quad \text{where} \quad \omega_m(t) = \frac{K_g v_c(t)}{r_{mp}} \quad (12)$$

We can now substitute the force function into equation (2):

$$M \dot{v}_c(t) + B_c v_c(t) = \frac{\eta_g K_g \eta_m k_t (V_m(t) r_{mp} - k_m K_g v_c(t))}{r_{mp}^2 R_m} \quad (13)$$

We can rearrange the equation to combine like-terms:

$$M \dot{v}_c(t) + \left(\frac{k_m \eta_g K_g^2 \eta_m k_t}{r_{mp}^2 R_m} + B_c \right) v_c(t) = \frac{\eta_g K_g \eta_m k_t V_m(t)}{r_{mp} R_m} \quad (14)$$

This equation can be simplified as follows:

$$M \dot{v}_c(t) + B_{eq} v_c(t) = A_m V_m(t) \quad (15)$$

where the equivalent damping term is:

$$B_{eq} = \frac{\eta_g K_g^2 \eta_m k_t k_m + B_c r_{mp}^2 R_m}{r_{mp}^2 R_m} \quad (16)$$

and the actuator gain is:

$$A_m = \frac{\eta_g K_g \eta_m k_t}{r_{mp} R_m} \quad (17)$$